MC68008 IN-CIRCUIT EMULATOR / BUS ANALYZER

Initial schematic | 48-pin CPU | ECP5-25F | Two independent 5 V bus interfaces

CPU socket + target plug

48-pin MC68008 socket / plug pin mapping
Independent CPU_* and TGT_* 5 V nets

File: cpu_sockets.kicad_sch

CPU receivers + drivers

Full CPU pin sensing and selected 5 V drive groups
Read data, responses and dedicated target clock

File: cpu_translation.kicad_sch

Target receivers + drivers

Full target pin sensing, including DMA activity
Separate master, data and auxiliary output enables

File: target_translation.kicad_sch

Output disable + open drain

Common default-disabled output-enable circuit
Four independently gated RESET / HALT low drivers

File: bus_safety.kicad_sch

USB / EEPROM / SPI / JTAG

FT2232H A = JTAG; B = asynchronous FT245 FIFO
EEPROM, shared SPI configuration flash, recovery header

File: usb_flash.kicad_sch

FPGA banks + power intent

ECP5-25F CABGA-256 bank and supply units
Perimeter-first I/O; target-powered rails and hardware interlock

File: fpga_core.kicad_sch

Target power / FPGA rails

Target +5V -> 3.3V buck / 1.1V buck / 2.5V LDO
Sequencing, independent voltage monitor and FPGA bypassing

File: power_supply.kicad_sch

SIGNAL CONVENTIONS
CPU_* / TGT_* = physical 5 V nets (never shorted together).
FPGA_CPU_*_IN / FPGA_TGT_*_IN = 3.3 V sensed signals.
FPGA_*_OUT = 3.3 V drive data; *_OE_N = active-low group enable.
Signal nets use global labels; supplies use power symbols. _N means active low.

DESIGN STILL TO COMPLETE
Validate assigned FPGA I/O, timing and BGA footprint.
Validate regulators, decoupling, hardware interlock and target power budget.
Finish FTDI support and boot straps; review target timing and ERC.
This is a schematic starting point; no PCB or fabrication release.

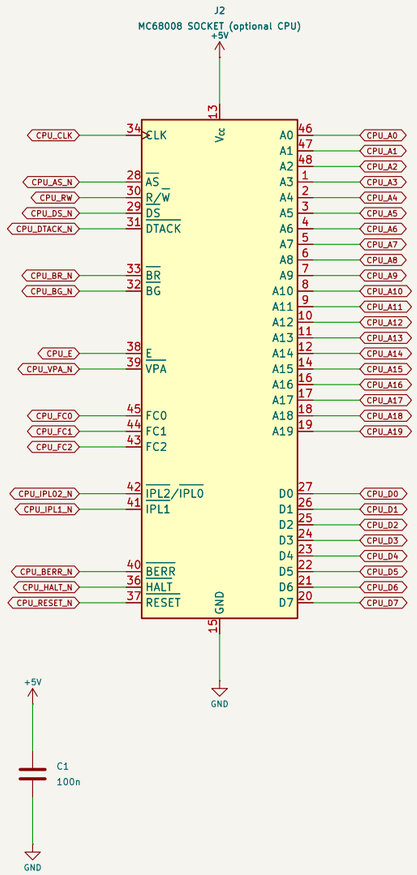
Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

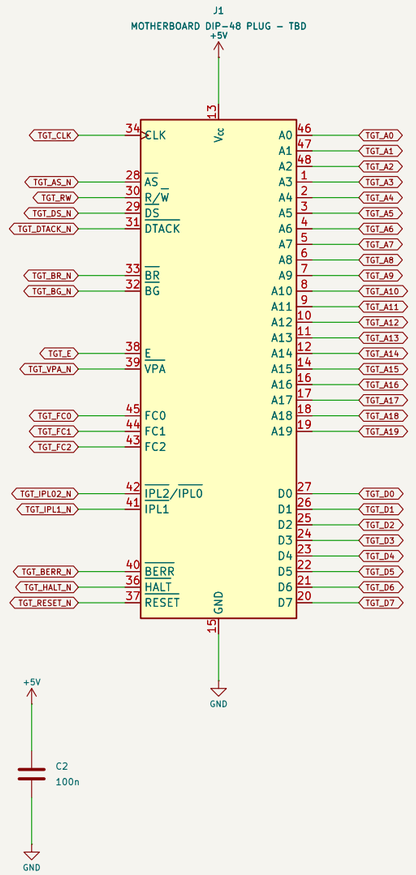
Sheet: /
File: 68008ice.kicad_sch

Title: MC68008 ICE — architecture and sheet index

Size: A2	Date: 2026-11-07	**Rev: 0.3 — POWER / USB-C**
KiCad E.D.A. 10.0.5		Id: 1/8

48-PIN MC68008: separate CPU and target domains





PINOUT NOTES
J1/J2 retain the MC68008P physical pin numbers. Socket symbols use passive pins.
A0–A19 and D0–D7: one DS strobe: IPL1 + combined IPL2/IPL0.
Two-wire BR/BG arbitration: no BGACK. No VMA output on this variant.
J1 plug footprint, bottom-view orientation and socket clearance remain to be selected.

POWER INTENT
+5V comes from motherboard pin 13 and powers
the physical CPU and AHCT bus drivers. USB VBUS stays separate.
Board power comes exclusively from this target rail; USB has no power path.

Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

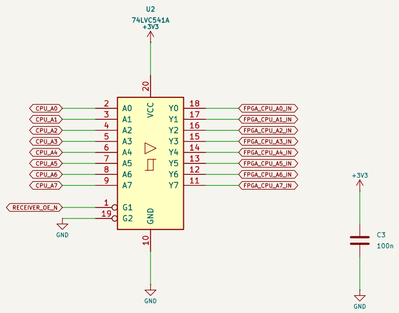
Sheet: /CPU socket + target plug/
File: cpu_sockets.kicad_sch

Title: 48-pin CPU socket and motherboard plug

Size: A3	Date: 2026-11-07	**Rev: 0.3 – POWER / USB-C**
KiCad E.D.A. 10.0.5		Id: 2/8

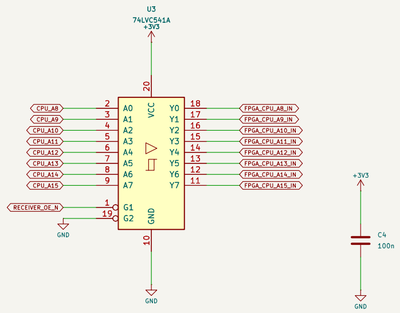
CPU INTERFACE | 5 V bus <–> dedicated 3.3 V FPGA signals

Receivers enable only after POWER_GOOD. AHCT drivers additionally require configuration valid and explicit arm. Forwarding remains combinational.



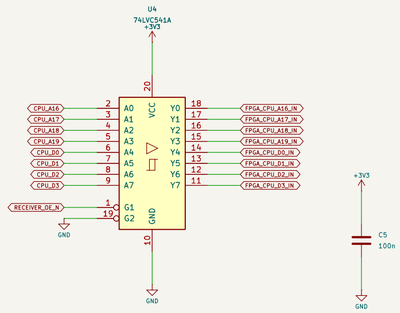
Sense A0, A1, A2, A3, A4, A5, A6, A7

FPGA balls ch0..7: R1, T2, T3, T4, B1, C1, D1, E1



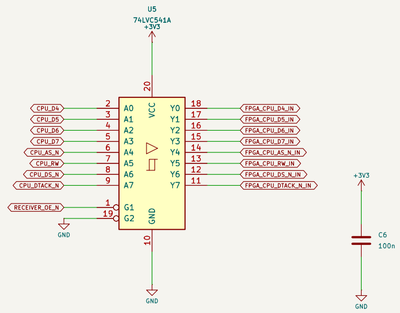
Sense A8, A9, A10, A11, A12, A13, A14, A15

FPGA balls ch0..7: F1, J1, K1, B3, B4, B5, B6, B7



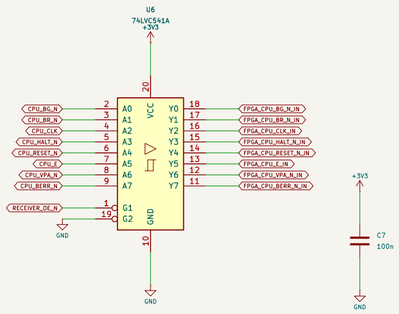
Sense A16, A17, A18, A19, D0, D1, D2, D3

FPGA balls ch0..7: L2, M2, P2, R2, H2, K2, C5, C7



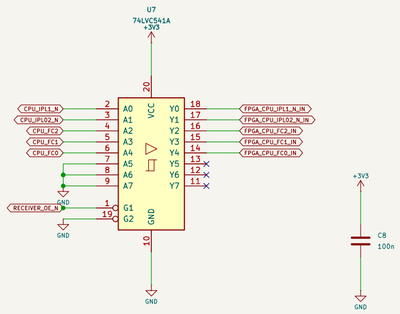
Sense D4, D5, D6, D7, AS_N, RW, DS_N, DTACK_N

FPGA balls ch0..7: M3, P3, P5, C3, R3, N5, E3, F3



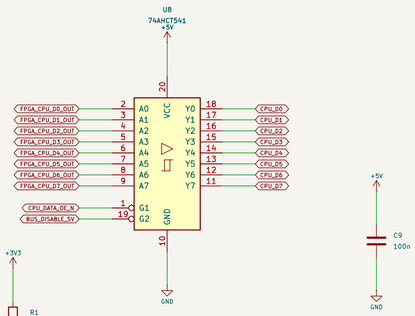
Sense BG_N, BR_N, CLK, HALT_N, RESET_N, E, VPA_N, BERR_N

FPGA balls ch0..7: B2, C2, F2, D6, N4, H3, N6, R4



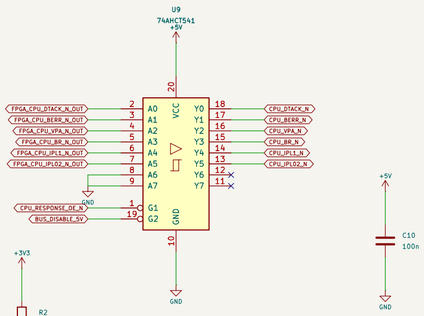
Sense IPL1_N, IPL02_N, FC2, FC1, FC0

FPGA balls ch0..7: D7, K4, D4, K3, J3, –, –, –



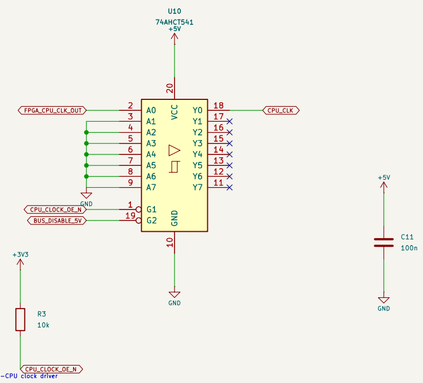
Read data / injected instruction bytes

FPGA balls ch0..7: J2, C4, C6, L3, N3, P4, P6, D3



FPGA independently selects each CPU response

FPGA balls ch0..7: G3, R5, F4, E2, E4, L4, –, –



Dedicated real-CPU clock driver

FPGA balls ch0..7: G2, –, –, –, –, –, –, –

OPEN ITEMS
All receiver/driver signals now terminate on U1. Ball references below each buffer show channel order.
Both sides sense every signal, including address during DMA: 6 receivers per side.
Driver grouping is provisional. Verify data/DTACK skew, clock duty cycle, OE turnaround and unpowered behavior.

Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

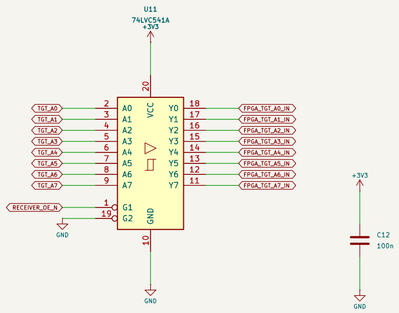
Sheet: /CPU receivers + drivers/
File: cpu_translation.kicad_sch

Title: Real CPU – sense and drive paths

Size: A2	Date: 2026-11-07	**Rev: 0.3 – POWER / USB-C**
KiCad E.D.A. 10.0.5		Id: 3/8

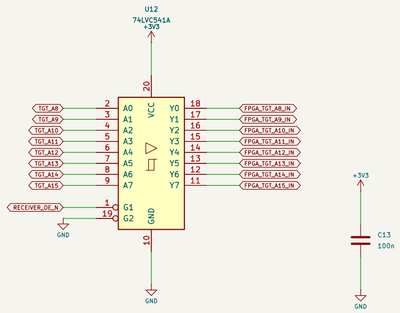
TGT INTERFACE | 5 V bus <–> dedicated 3.3 V FPGA signals

Receivers enable only after POWER_GOOD. AHCT drivers additionally require configuration valid and explicit arm. Forwarding remains combinational.



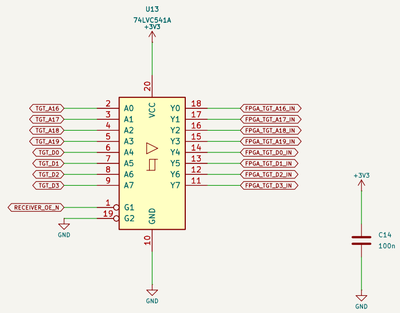
Sense A0, A1, A2, A3, A4, A5, A6, A7

FPGA balls ch0..7: B16, C16, D16, E16, F16, G16, K16, M16



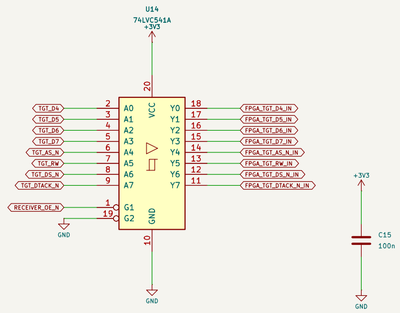
Sense A8, A9, A10, A11, A12, A13, A14, A15

FPGA balls ch0..7: P16, T13, T15, B9, B11, B13, B15, E15



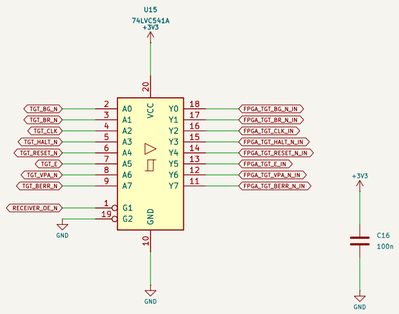
Sense A16, A17, A18, A19, D0, D1, D2, D3

FPGA balls ch0..7: G15, J15, L15, P15, C11, C13, D14, F14



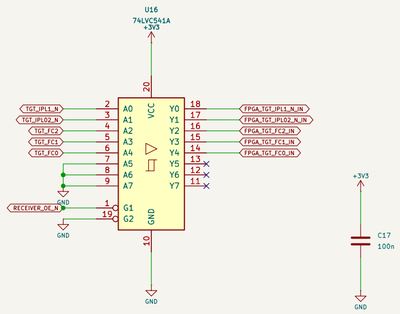
Sense D4, D5, D6, D7, AS_N, RW, DS_N, DTACK_N

FPGA balls ch0..7: H14, K14, M14, P11, R13, N12, P13, D8



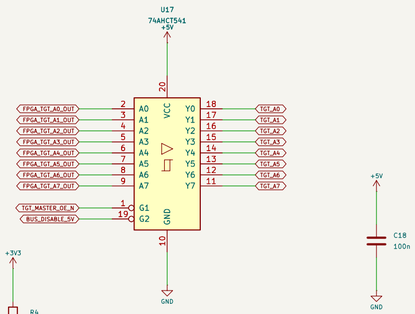
Sense BG_N, BR_N, CLK, HALT_N, RESET_N, E, VPA_N, BERR_N

FPGA balls ch0..7: C8, C10, A7, J13, N11, D9, G4, R15



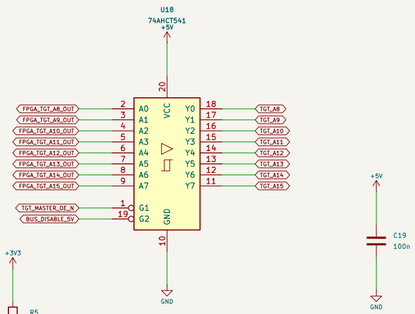
Sense IPL1_N, IPL02_N, FC2, FC1, FC0

FPGA balls ch0..7: K13, L13, F13, D13, D11, –, –, –



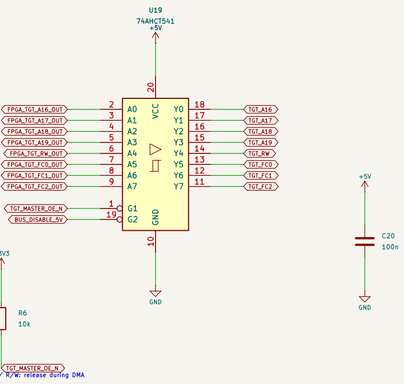
Address low: release during DMA

FPGA balls ch0..7: T6, R6, R7, P7, N7, J16, L16, N16



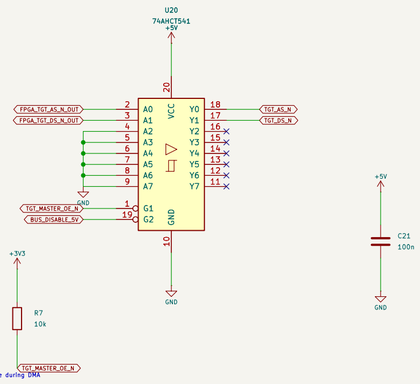
Address middle: release during DMA

FPGA balls ch0..7: R16, T14, B8, B10, B12, B14, C15, F15



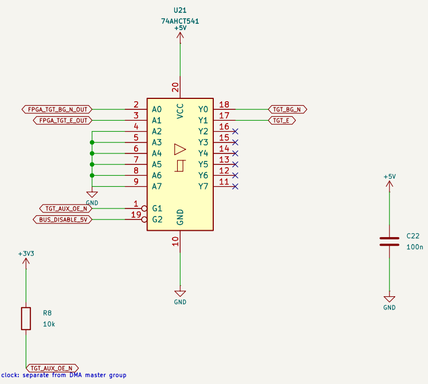
Address / function / R/W: release during DMA

FPGA balls ch0..7: H15, K15, M15, R12, N13, D12, E13, G13



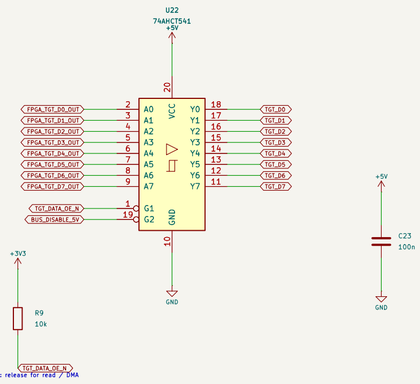
Strobes: release during DMA

FPGA balls ch0..7: R14, P14, –, –, –, –, –, –



Grant and E clock: separate from DMA master group

FPGA balls ch0..7: C9, D10, –, –, –, –, –, –



Write data only: release for read / DMA

FPGA balls ch0..7: C12, C14, E14, G14, J14, L14, N14, P12

OPEN ITEMS
All receiver/driver signals now terminate on U1. Ball references below each buffer show channel order.
Both sides sense every signal, including address during DMA: 6 receivers per side.
Driver grouping is provisional. Verify data/DTACK skew, clock duty cycle, OE turnaround and unpowered behavior.

Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

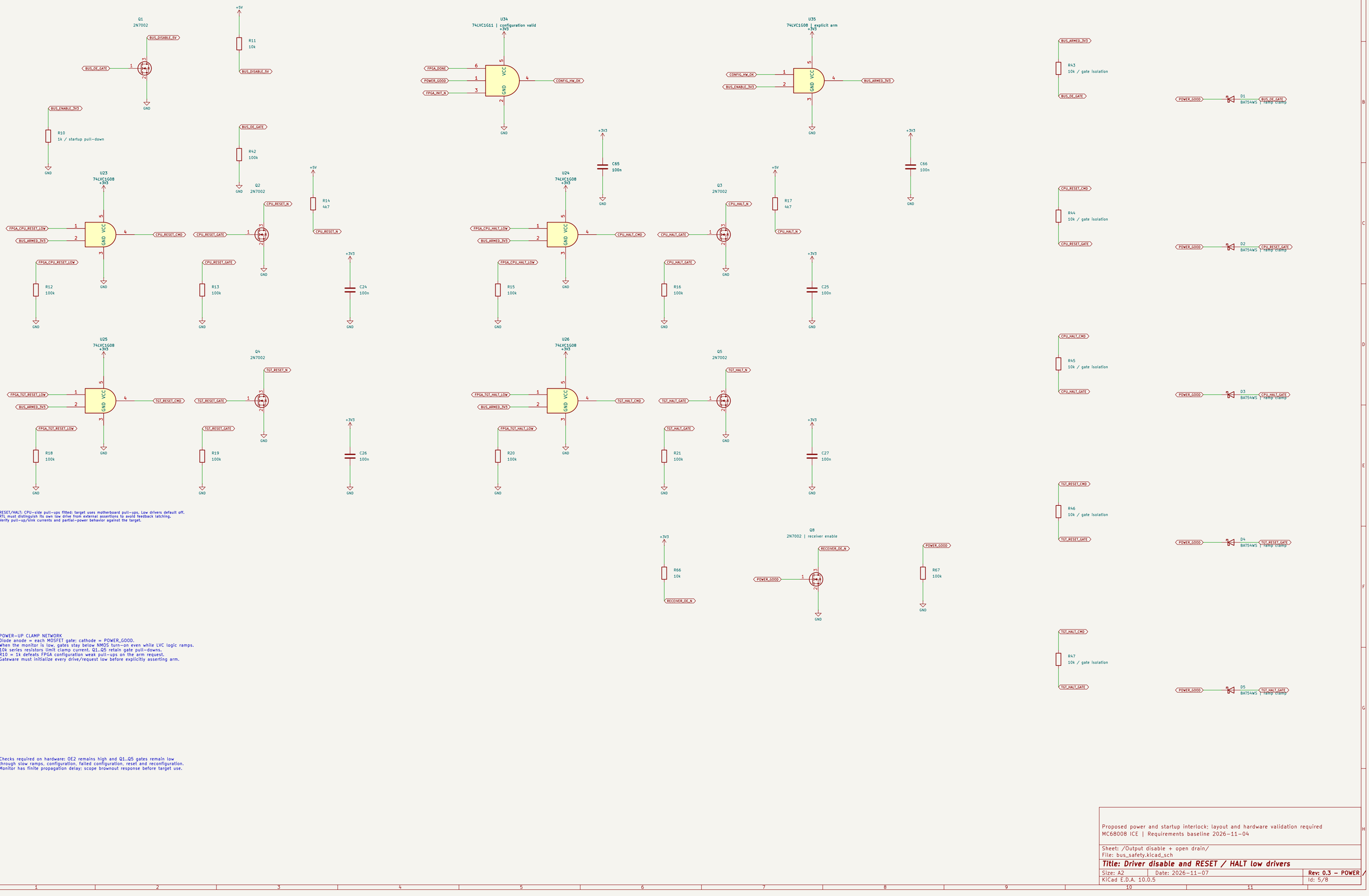
Sheet: /Target receivers + drivers/
File: target_translation.kicad_sch

Title: Motherboard – sense and drive paths

Size: A2	Date: 2026-11-07	**Rev: 0.3 – POWER / USB-C**
KiCad E.D.A. 10.0.5		Id: 4/8

BUS OUTPUT SAFETY / RESET AND HALT

ARMED = REQUEST & POWER_GOOD & FPGA_DONE & FPGA_INIT_N
AHCT /OE2 remains pulled up to target +5V until ARMED. Hardware clamps keep every low-driver gate off during supply ramp.



RESET/HALT: CPU-side pull-ups fitted; target uses motherboard pull-ups. Low drivers default off.
RTL must distinguish its own low drive from external assertions to avoid feedback latching.
Verify pull-up/sink currents and partial-power behavior against the target.

POWER-UP CLAMP NETWORK
Diode anode = each MOSFET gate; cathode = POWER_GOOD.
When the monitor is low, gates stay below NMOS turn-on even while LVC logic ramps.
10k series resistors limit clamp current. Q1..Q5 retain gate pull-downs.
R10 = 1k defeats FPGA configuration weak pull-ups on the arm request.
Gateware must initialize every drive/request low before explicitly asserting arm.

Checks required on hardware: OE2 remains high and Q1..Q5 gates remain low
through slow ramps, configuration, failed configuration, reset and reconfiguration.
Monitor has finite propagation delay; scope brownout response before target use.

Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

Sheet: /Output disable + open drain/
File: bus_safety.kicad_sch

Title: Driver disable and RESET / HALT low drivers

Size: A2	Date: 2026-11-07	**Rev: 0.3 – POWER / USB-C**
KiCad E.D.A. 10.0.5		Id: 5/8

HOST / CONFIGURATION | signal allocation started, support circuitry incomplete

U27
FT2232HL

CHANNEL A: MPSSE / JTAG
ADBUS0=TCK, 1=TDI, 2=TDO, 3=TMS.
Remove JTAG source links for external programming.

CHANNEL B: asynchronous FT245
BDBUS0..7 = D0..7
BCBUS0=RXF#, 1=TXE#, 2=RD#, 3=WR#, 4=SIWU#.
EEPROM must select channel B FIFO mode.

J3
JTAG RECOVERY (custom pinout)

Header is 3.3 V; documented custom pinout.
Disconnect FTDI TCK/TDI/TMS links before external drive.

#FLG5
PWR_FLAG

U28
93LC46B (x16) — provisional

#FLG6
PWR_FLAG

Q6
2N7002 | VBUS sense

Shared FPGA boot / application SPI flash.
FPGA user logic must retain access to the config-SPI pins.
Flash size and image directory offsets are provisional.
JTAG remains the independent recovery path.

U29
W25Q32JVSS (4 MiB) — provisional

J4
USB-C | target-powered USB2 device

U36
TPD2EUSB30 | rail-less USB ESD

NEORV32 APPLICATION DEBUG (planned gateware)
FT2232H A -> ECP5 configuration JTAG -> JTAGG bridge -> NEORV32 OCD.
Enable OCD_EN; tunnel the JTAG TAP or implement an external DTM.
Requires matching OpenOCD configuration; no extra debug IC or GPIO is required.
Keep the bus-safety logic outside the NEORV32 debug/reset domain.

SUPPORT CIRCUITRY TO COMPLETE
FTDI remaining: 12 MHz crystal/load caps, final VCCIO bypassing and EEPROM programming.
EEPROM organization, resistor network and programming setup need final datasheet review. USB differential routing is a PCB task.
FTDI power and clock labels are unfinished connection points. No regulators or complete USB support circuit is implied.

USB-C: one 5.1k Rd per CC pin. Both USB2 D+/D− pairs are joined.
VBUS connects only to Q6 gate sensing through 100k, with a 1M bleed.
All FTDI supplies derive from target power; no VBUS-to-supply path.
SBU pins NC; shell bonded to GND. No PD or SuperSpeed circuitry.

Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

Sheet: /USB / EEPROM / SPI / JTAG/
File: usb_flash.kicad_sch

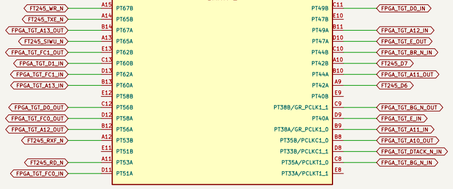
Title: FT2232H host link, EEPROM, flash and JTAG

Size: A2 | Date: 2026-11-07 | **Rev: 0.3 — POWER / USB-C**
KiCad E.D.A. 10.0.5 | Id: 6/8

U1A
LFE5U-25F CABGA256



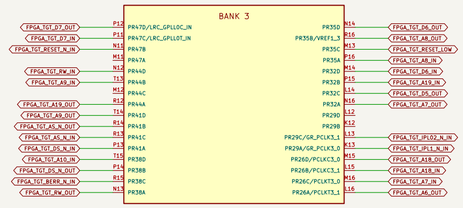
U1B
LFE5U-25F CABGA256



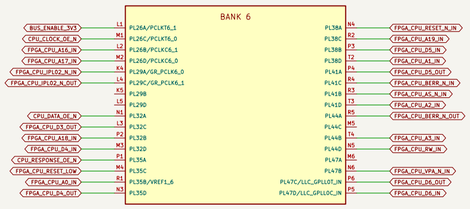
U1C
LFE5U-25F CABGA256



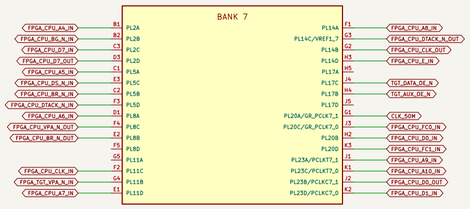
U1D
LFE5U-25F CABGA256



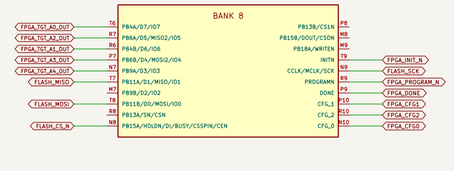
U1E
LFE5U-25F CABGA256



U1F
LFE5U-25F CABGA256



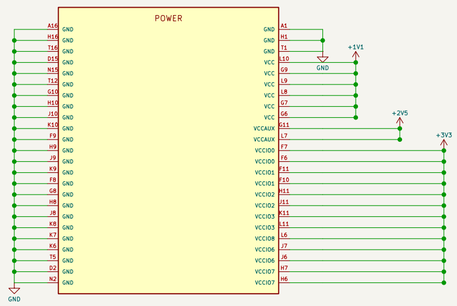
U1G
LFE5U-25F CABGA256



U1H
LFE5U-25F CABGA256



U1I
LFE5U-25F CABGA256















PERIMETER-FIRST PIN ALLOCATION
All 152 bus signals and 13 FT245 signals connect to U1; CLK_50M is reserved for the oscillator.
A7 = target clock (PCLKT0_0); G1 = sampling clock (GR_PCLK7_1).
Use outer BGA rows/columns before inner rings, subject to pin-function and boot constraints.
See fpga_pins.csv and 68008ice.lpf for assignments and remaining spare pins.

POWER / BOOT DESIGN STILL REQUIRED
Nominal rails: VCC 1.1 V, VCCAUX 2.5 V, VCCIO 3.3 V (verify speed-grade limits).
Regulators, sequencing, monitor and baseline decoupling are on the power sheet.
CFG straps select master-SPI 010; PROGRAMN / INITN / DONE have pull-ups.
Hardware qualifies the arm request with power-good, DONE and INITN.
Validate the CABGA-256 footprint from package drawing; footprint deliberately unset.

BANK 8 / BOOT
Master-SPI flash CS = N8 (CSSPIN), not the slave CSN pin R8.
D2..D7 may serve user outputs after single-SPI configuration.
Release unused config ports in gateware; keep all bus drivers off until configured.
CCLK user access requires USRMCLK: fixed JTAG pins keep their dedicated functions.

Proposed power and startup interlock; layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

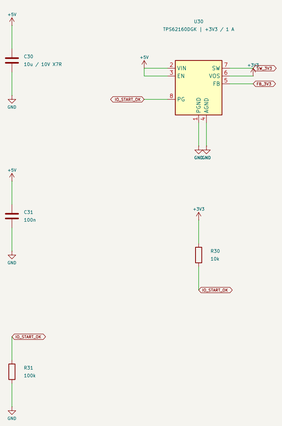
Sheet: /FPGA banks + power intent/
File: fpga_core.kicad_sch

Title: ECP5-25F CABGA-256 — banks, configuration and power

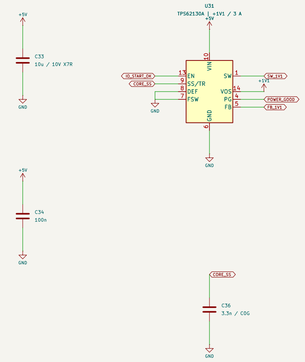
Size: A2	Date: 2026-11-07	**Rev: 0.3 — POWER / USB-C**
KiCad E.D.A. 10.0.5		Id: 7/8

TARGET POWER ONLY | 3.3 V FIRST → 1.1 V CORE + 2.5 V AUX

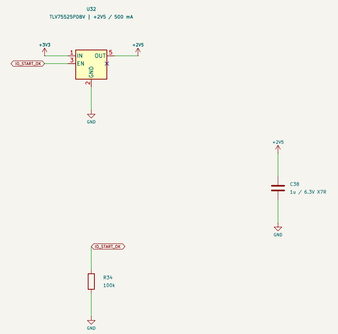
Regulator current ratings provide headroom; actual FPGA current requires a post-synthesis power estimate. No USB power input.



Vout = 0.8 * (1 + 312k/100k) = 3.296 V
PG enables the core buck and AUX LDO only after VCCIOB/flash power is established.
Kelvin-connect VOS and divider to the output capacitor. Keep SW copper small.

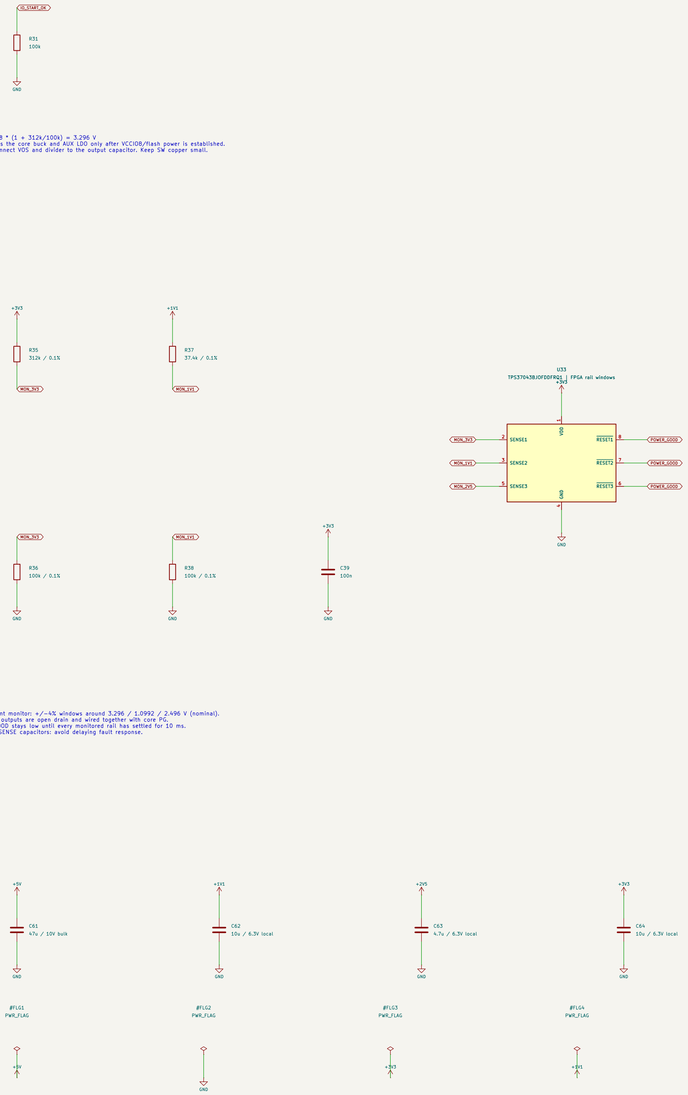


Vout = 0.8 * (1 + 37.4k/100k) = 1.0992 V
FSW = low (2.5 MHz); DEF = low; SS/TR sets a controlled ramp.
Core PG is wired into the independent power-good interlock.



AUX derives from 3.3 V, not directly from 5 V.
At 100 mA: LDO dissipation is approximately 80 mW.
500 mA is the electrical rating; check package temperature at higher load.

Independent monitor: +/−4% windows around 3.296 / 1.0992 / 2.496 V (nominal).
All RESET outputs are open drain and wired together with core PG.
POWER_GOOD stays low until every monitored rail has settled for 10 ms.
No large SENSE capacitors: avoid delaying fault response.



Only supply source: motherboard J1 pin 13 (+5V), return pins 15 and 35.
Verify socket-contact, track and target-PSU current capacity using a measured power budget.
Rated regulator currents are not a prediction of total target input current.

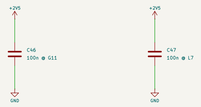
Target +5V supervisor: 4.65 V nominal falling threshold, CT open = 20 ms release.
Allow for threshold tolerance and hysteresis: worst-case startup may need about 4.86 V.
Its open-drain RESET also clamps POWER_GOOD: AHCT drive cannot arm on a slow low-voltage ramp.

FPGA LOCAL DECOUPLING | Place at the indicated balls / banks; short ground vias and solid planes

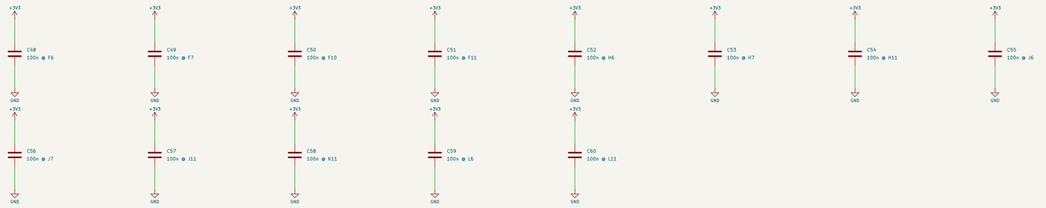
+1V1 : G6, G7, G9, L8, L9, L10



+2V5 : G11, L7



+3V3 : F6, F7, F10, F11, H6, H7, H11, J6, J7, J11, K11, L6, L11



Baseline: one 100 nF at each VCC/VCCAUX/VCCIO ball.
Final values and count require PCB placement and PDN review.

Proposed power and startup interlock: layout and hardware validation required
MC68008 ICE | Requirements baseline 2026-11-04

Sheets: /Target power / FPGA rails/
File: power_supply.kicad_sch

Title: Target-powered FPGA rails, sequencing and decoupling

Size: A1	Date: 2026-11-07	Rev: 0.3 – POWER	USB-C
KiCad E.D.A. 10.0.5		Id: 8/8	